

METEOROLOGY FORCINGS

GLY606 Water Data Analysis & Modeling

Nov 4th 2024





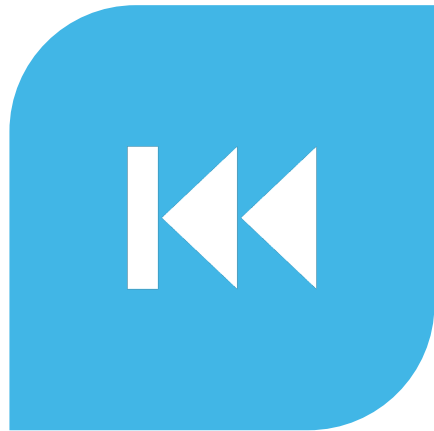
git

≠

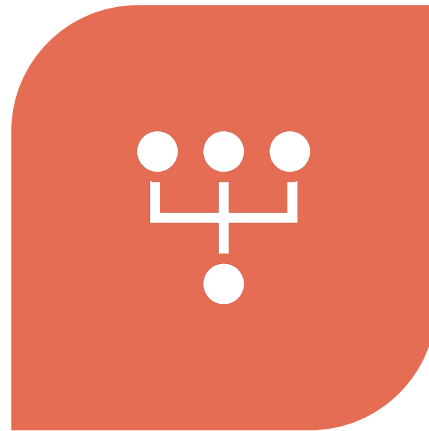


GitHub

Three use cases



#1: HISTORY OF
VERSIONS



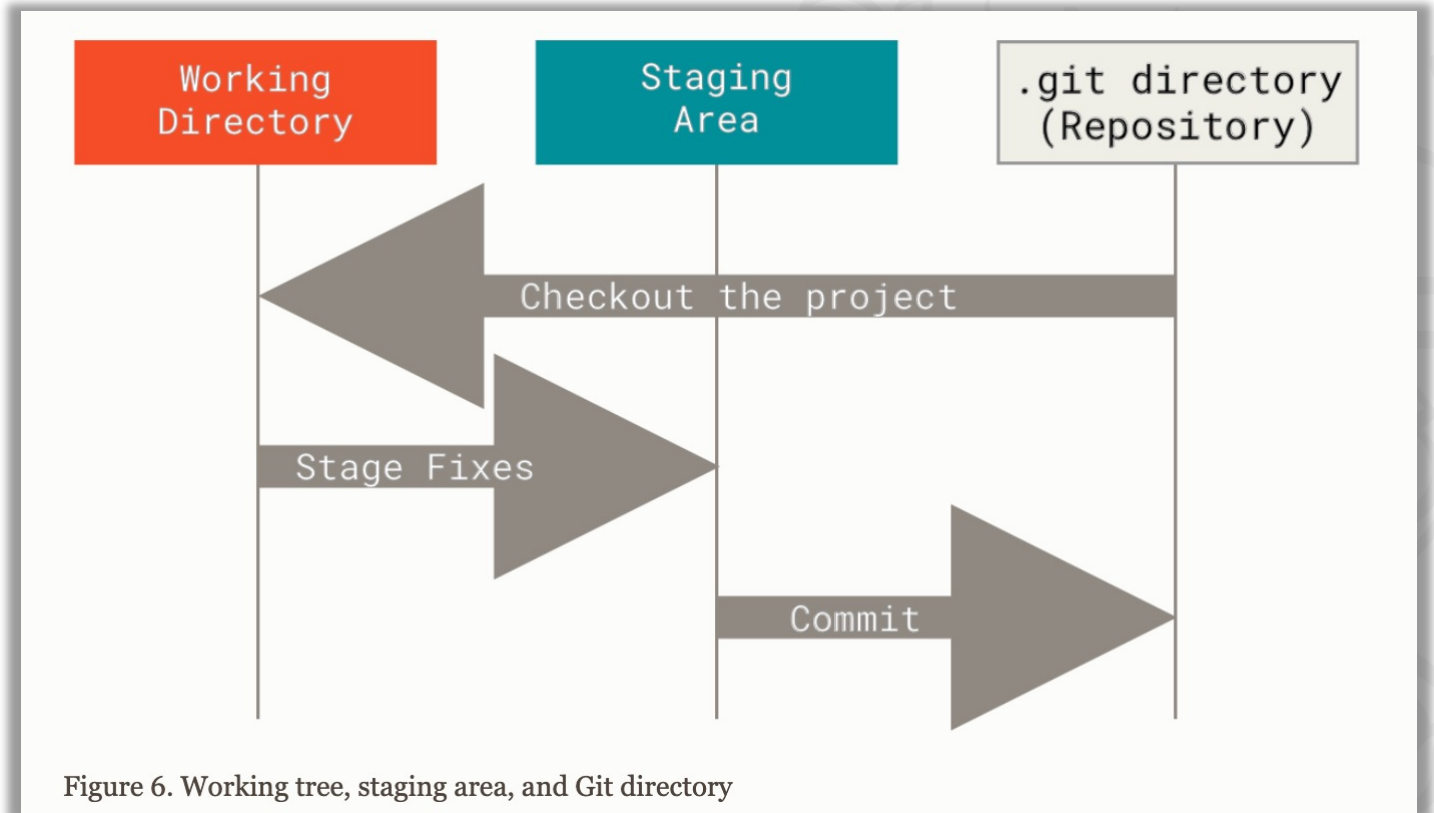
#2: BRANCHING



#3: COLLABORATION
(GITHUB)

Use case #1: history of versions

- Basic workflow
- **Adding** and **committing** files
- The git log
- The **staging** area
- **Removing** files
- Viewing diffs of files
- The .gitignore file

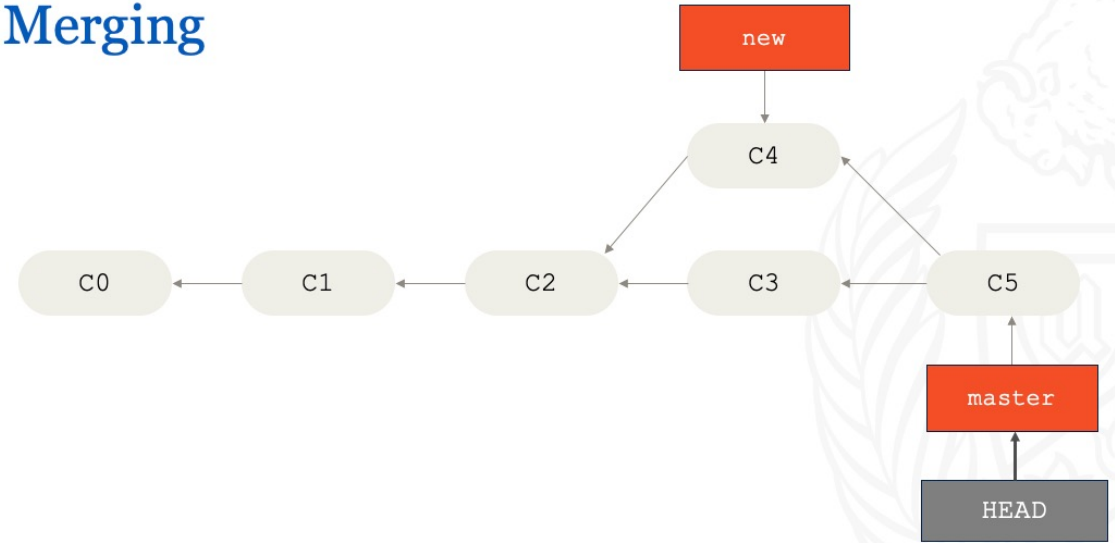


Source: <https://git-scm.com/book/en/v2/Getting-Started-What-is-Git%3F#:~:text=Git%20has%20three%20main%20states,into%20your%20next%20commit%20snapshot.>

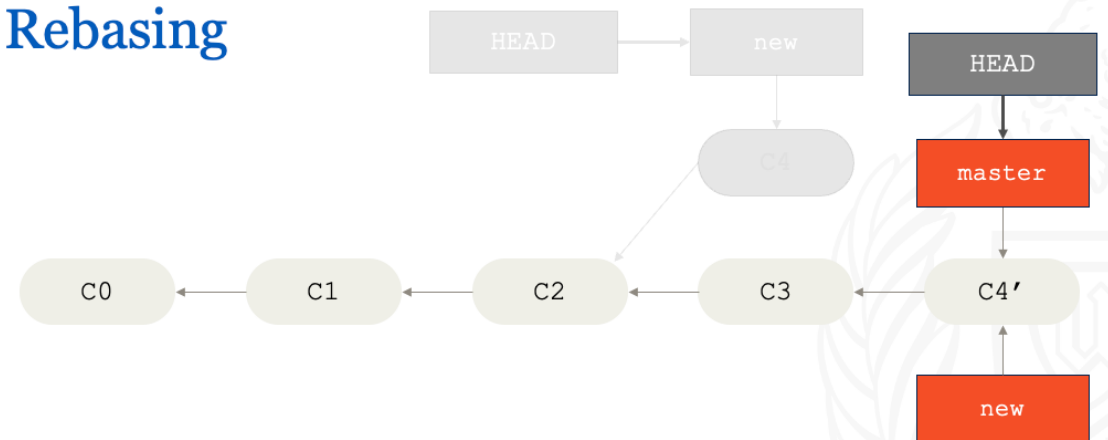
Use case #2: branching

- What is a branch?
- **Branching** commands
- The HEAD pointer
- Basics of **merging**
- Basics of **rebasing**
- Aside: the git reset command

Merging

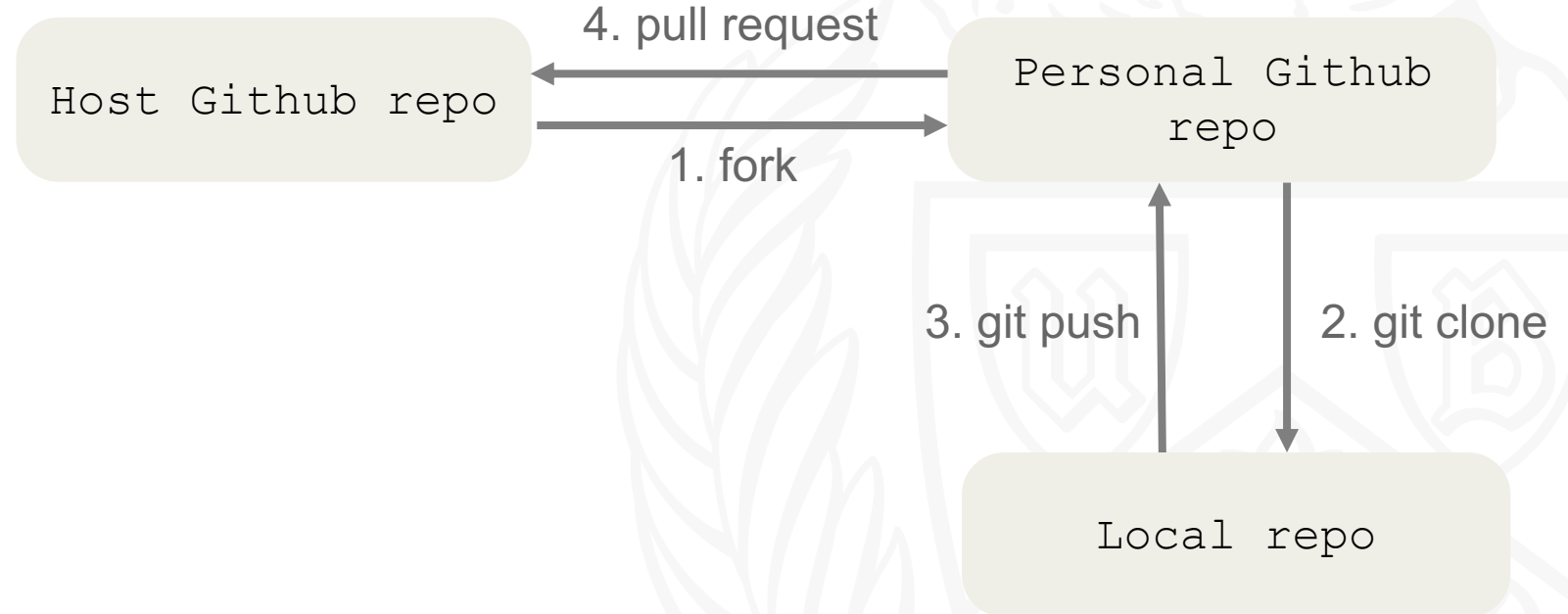


Rebasing

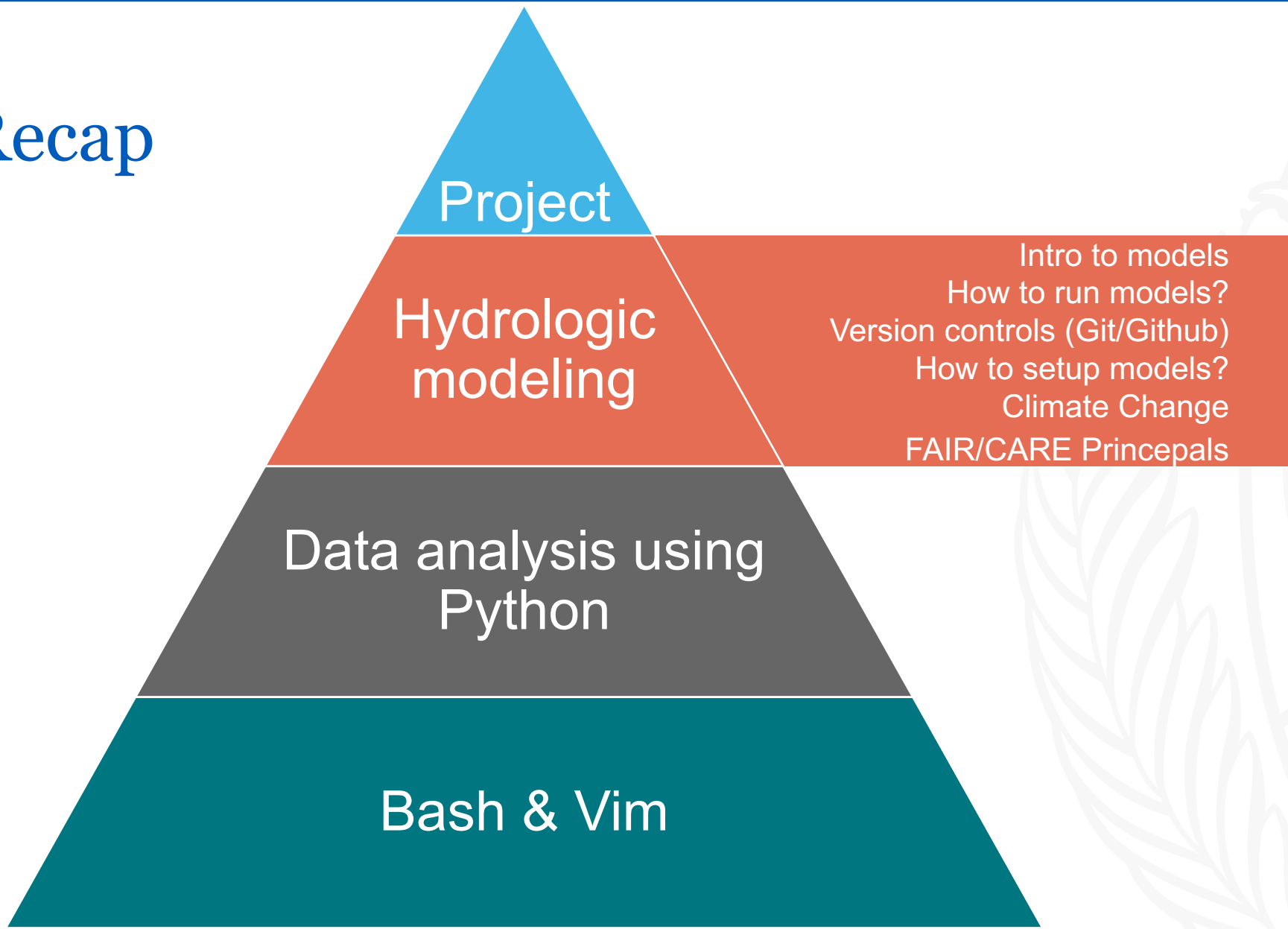


Use case #3: collaboration

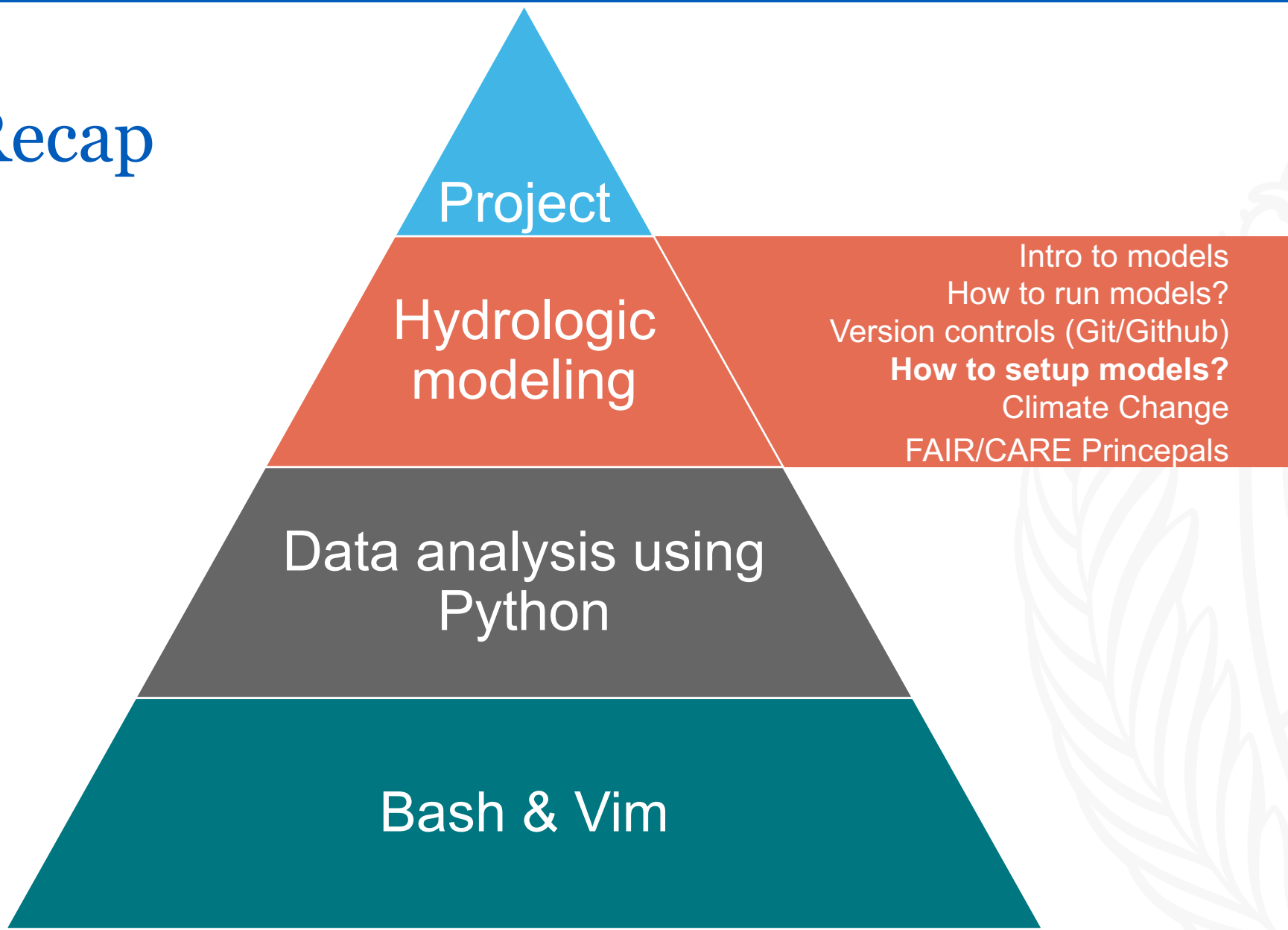
- Creating a repo on Github
- **Git clone**
- Remotes
- Remote-tracking branches
- Push, fetch, and pull



Recap



Recap



Meteorological forcing data

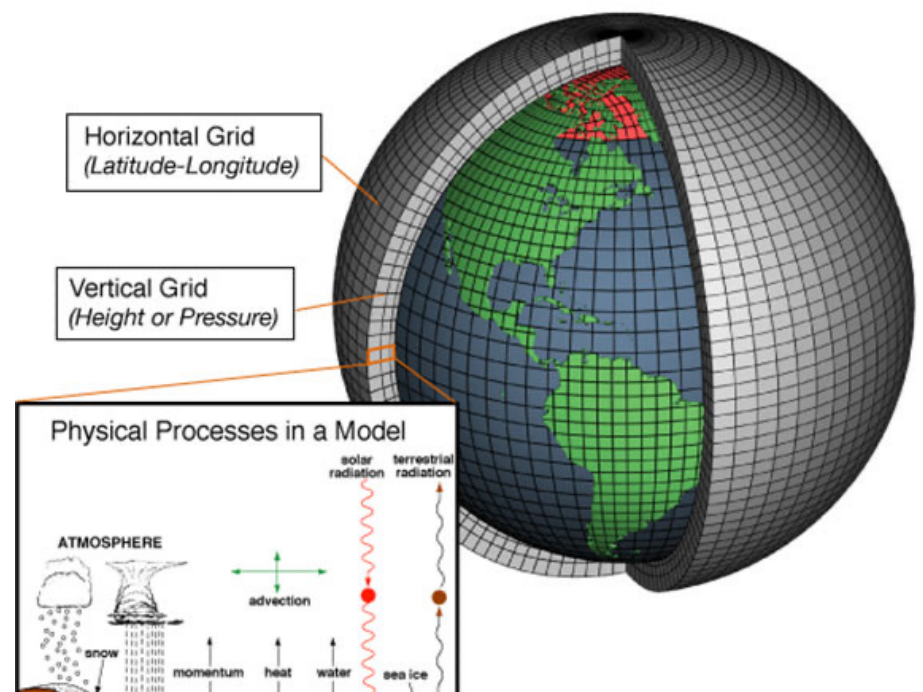
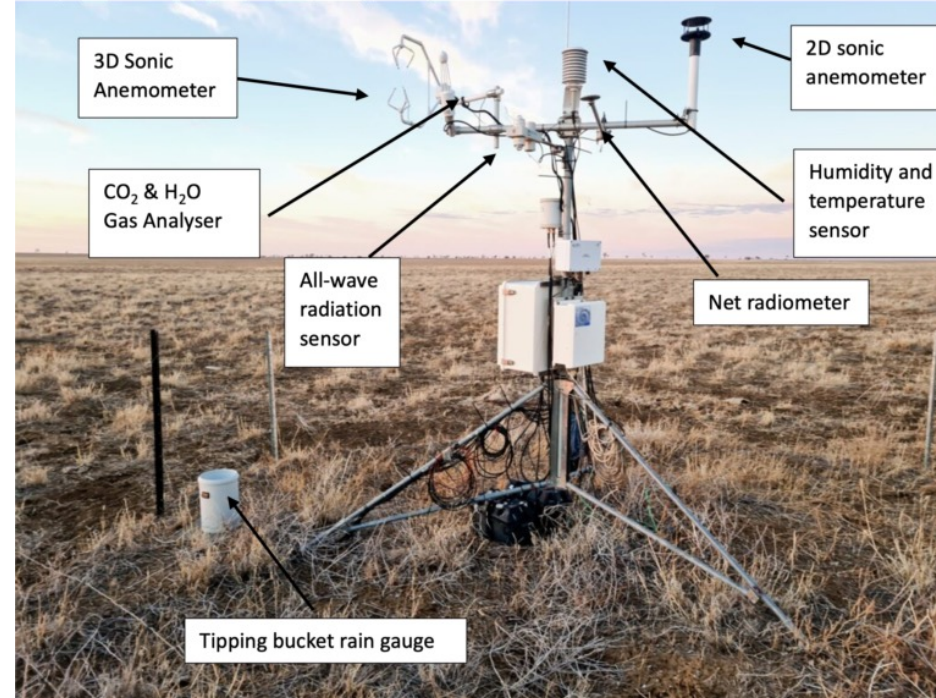
- A set of **weather variables** like temperature, precipitation, humidity, wind speed, air pressure, etc., which are used as input data to drive hydrological models

Meteorological Forcings, Required in all simulations:

Variable	Description	Units
AIR_TEMP	Average air temperature	C
PREC	Total precipitation (rain and snow)	mm
PRESSURE	Atmospheric pressure	kPa
SWDOWN	Incoming shortwave radiation	W/m ²
LWDOWN	Incoming longwave radiation	W/m ²
VP	Vapor pressure	kPa
WIND	Wind speed	m/s

mm/timestep

This value would be different for 3-hourly versus daily timesteps.



Source of meteorological data

- Onsite observations
- Remote sensing data
- Atmospheric models

AmeriFlux

- AmeriFlux is a network of PI-managed sites measuring ecosystem CO₂, water, and energy fluxes in North, Central and South America.
- It usually provides measurements of all necessary variables for hydrologic modeling
 - Extra quality control might be required

<https://ameriflux.lbl.gov/about/about-ameriflux/>

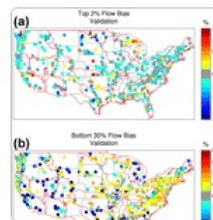


Large sample watershed-scale dataset for the contiguous USA

Research article | Highlight paper | 

14 Jan 2015

Development of a large-sample watershed-scale hydrometeorological data set for the contiguous USA: data set characteristics and assessment of regional variability in hydrologic model performance



A. J. Newman , M. P. Clark, K. Sampson, A. Wood, L. E. Hay, A. Bock, R. J. Viger, D. Blodgett, L. Brekke, J. R. Arnold, T. Hopson, and Q. Duan

Research article | Highlight paper | 

20 Oct 2017

The CAMELS data set: catchment attributes and meteorology for large-sample studies



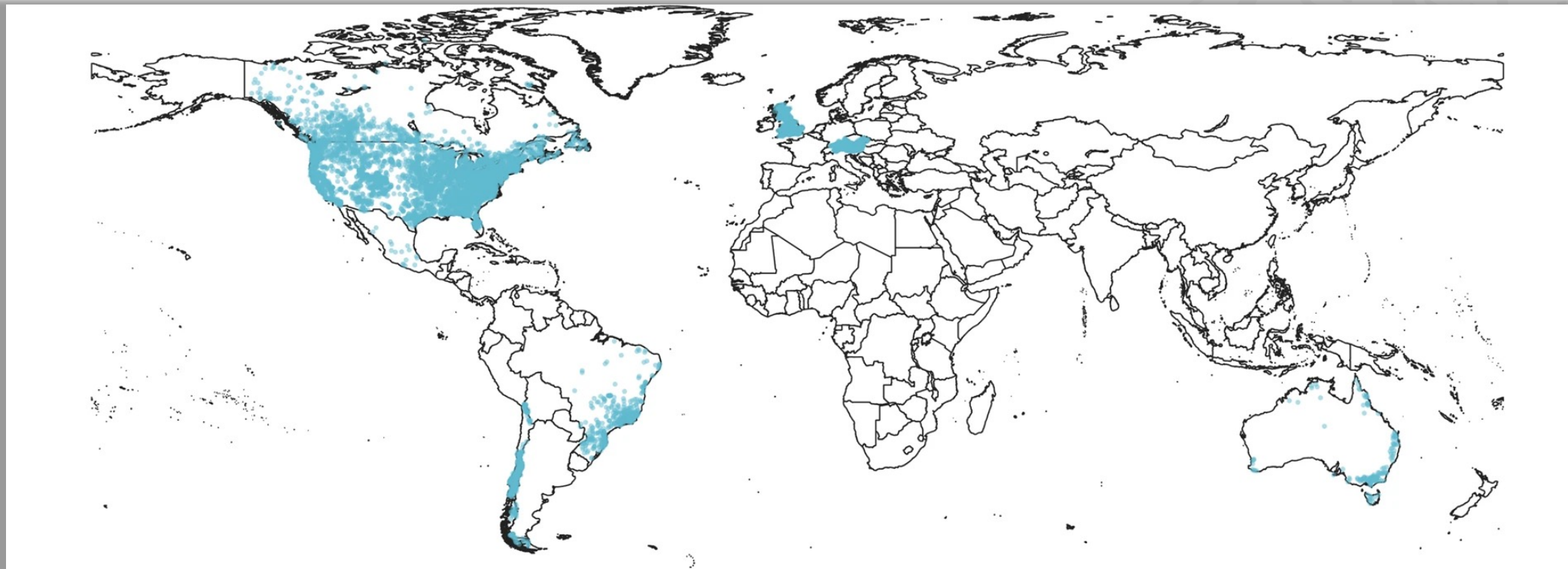
Nans Addor , Andrew J. Newman, Naoki Mizukami, and Martyn P. Clark

Large sample watershed-scale dataset globally

Data Descriptor | [Open access](#) | Published: 31 January 2023

Caravan - A global community dataset for large-sample hydrology

[Frederik Kratzert](#) , [Grey Nearing](#), [Nans Addor](#), [Tyler Erickson](#), [Martin Gauch](#), [Oren Gilon](#), [Lukas Gudmundsson](#), [Avinatan Hassidim](#), [Daniel Klotz](#), [Sella Nevo](#), [Guy Shalev](#) & [Yossi Matias](#)



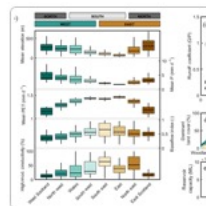
Large sample watershed-scale dataset globally

Data description paper | 

12 Oct 2020

CAMELS-GB: hydrometeorological time series and landscape attributes for 671 catchments in Great Britain

Gemma Coxon , Nans Addor, John P. Bloomfield, Jim Freer, Matt Fry, Jamie Hannaford, Nicholas J. K. Howden, Rosanna Lane, Melinda Lewis, Emma L. Robinson, Thorsten Wagener, and Ross Woods

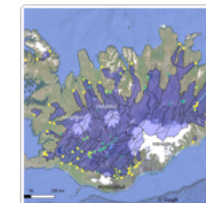


Data description paper | 

13 Jun 2024

LamaH-Ice: Large-SaMple DAta for Hydrology and Environmental Sciences for Iceland

Hordur Bragi Helgason  and Bart Nijssen

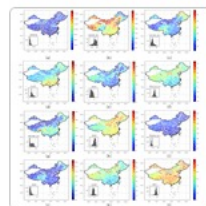


Data description paper | 

03 Dec 2021

CCAM: China Catchment Attributes and Meteorology dataset

Zhen Hao , Jin Jin , Runliang Xia, Shimin Tian, Wushuang Yang, Qixing Liu, Min Zhu, Tao Ma, Chengran Jing, and Yanning Zhang

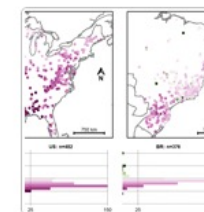


Research article | Highlight paper | 

12 Sep 2024

Large-sample hydrology – a few camels or a whole caravan?

Franziska Clerc-Schwarzenbach , Giovanni Selleri, Mattia Neri, Elena Toth, Ilja van Meerveld, and Jan Seibert



What do you think?

1846

Grid-based meteorological datasets (operational)

- ERA5

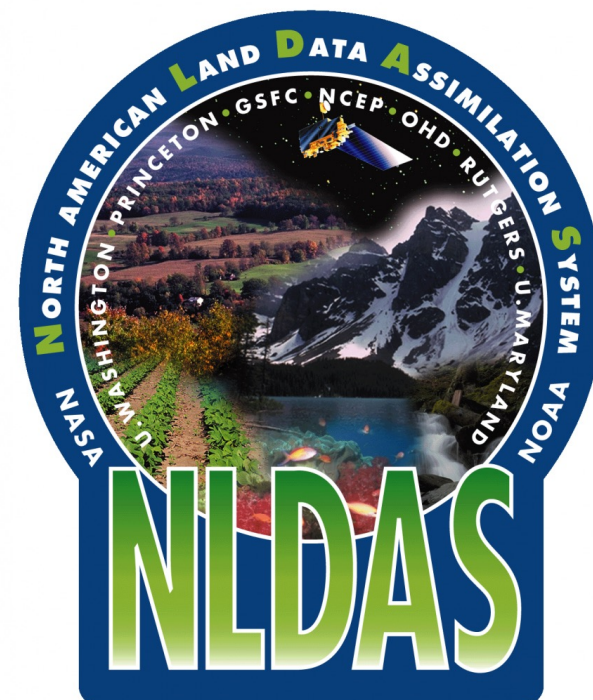
**31km
Hourly**

- <https://www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5>
- ERA5 is the fifth generation ECMWF reanalysis for the global climate and weather for the past 8 decades.

- GLDAS/NLDAS

**0.125°
Hourly**

- <https://ldas.gsfc.nasa.gov/nldas>
- North American Land Data Assimilation System (NLDAS) aims to construct quality-controlled, and spatially and temporally consistent, land-surface model (LSM) datasets from the best available observations and reanalyses to support modeling activities.



Grid-based meteorological datasets

- Daymet (<https://daymet.ornl.gov/>)
 - Daymet provides long-term, continuous, gridded estimates of daily weather and climatology variables by interpolating and extrapolating ground-based observations through statistical modeling techniques.
- GMET (<https://ncar.github.io/hydrology/models/GMET>)
 - Gridded Meteorological Ensemble Tool (GMET) allows for quantification of uncertainty for station-based gridded precipitation and temperature datasets
- gridMET (<https://www.climatologylab.org/gridmet.html>)
 - gridMET is a dataset of daily surface meteorological data covering the contiguous US from 1979-yesterday.

Compared to ERA5 and NLDAS, what are the major differences for these datasets?

Grid-based meteorological datasets

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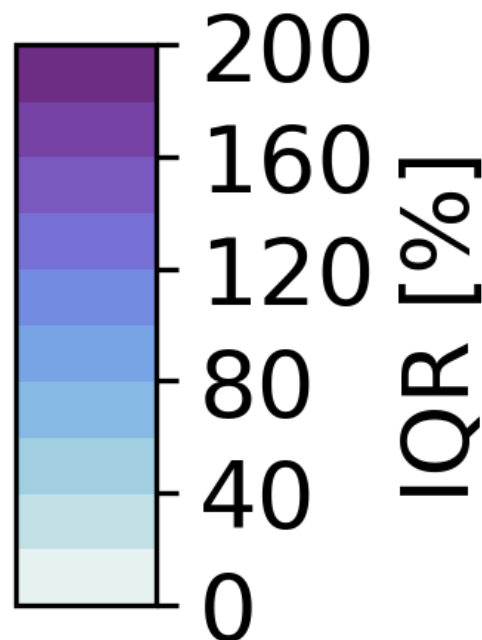
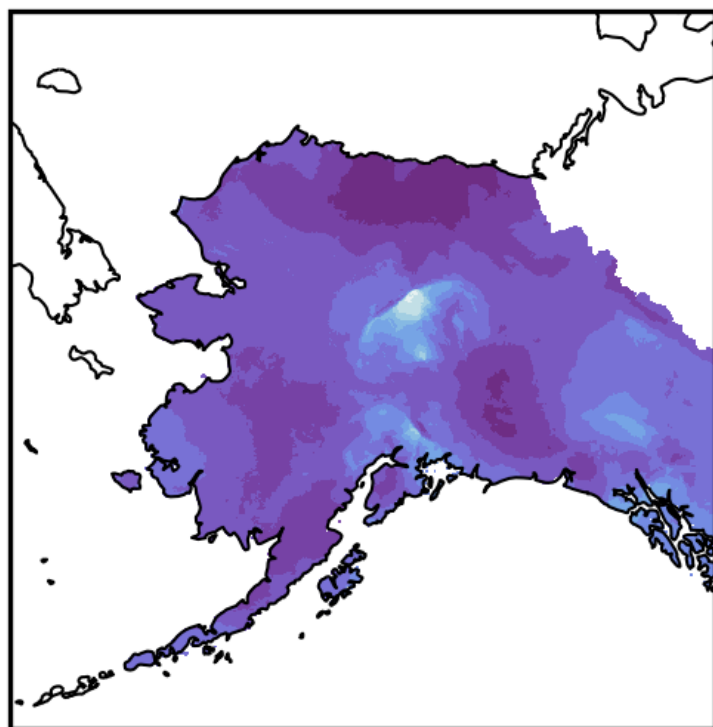
High resolution
(1km)

Uncertainty
Quantification

Semi-operational
(Downscaled NLDAS)

Biases exist in the gridded meteorological forcings

DJF



Example: Relative Interquartile Range for GMET data in Alaska

$$IQR = \frac{P_{75\%} - P_{25\%}}{(P_{75\%} + P_{25\%})/2}$$

Biases in the met forcings are major sources for biases in hydrologic simulations

Example: Steps to process meteorological forcings from NLDAS

- Step 1: Download the datasets
- Step 2: Concatenate the hourly files into the targeted format
- Step 3: Convert the units of the variables to the units required by the VIC model
- Step 4: If necessary, regridding the met forcing data to the targeted domain grid

Live Demo

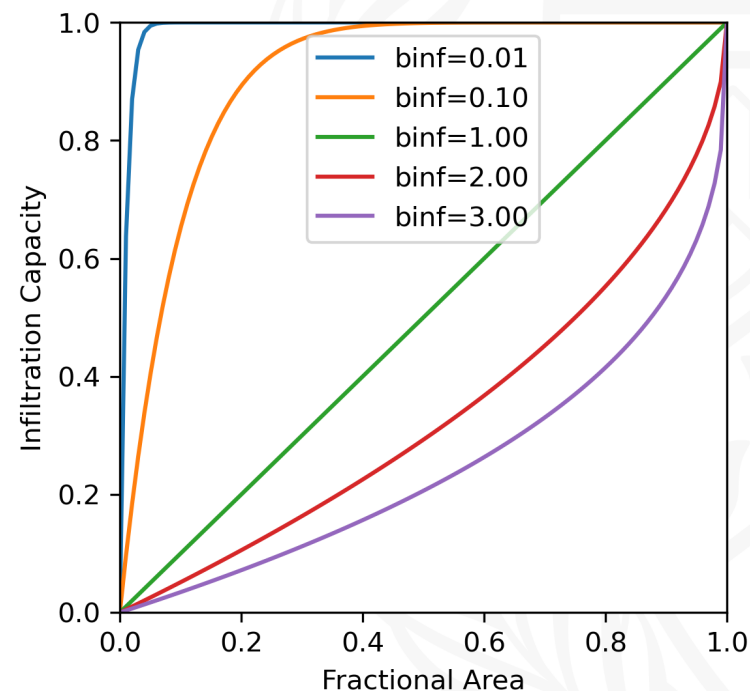
<https://ldas.gsfc.nasa.gov/nldas>

General rules when preparing meteorological forcings...

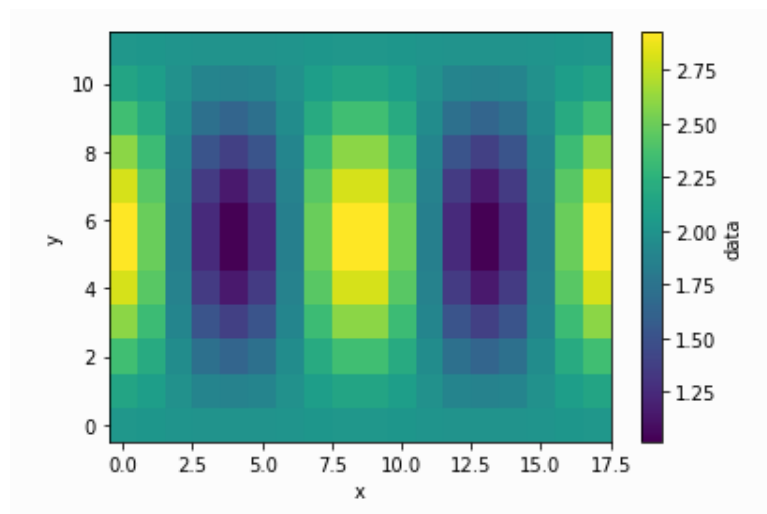
When the spatial resolutions of land surface/parameter datasets and meteorological forcing are different, it is recommended to **regrid the met forcings**.

20%clay 80% silt $B_{inf}=0.1$	60% clay 40% silt $B_{inf}=1$
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The arithmetic average works when calculating soil composition but it won't work when calculating non-linear parameters!



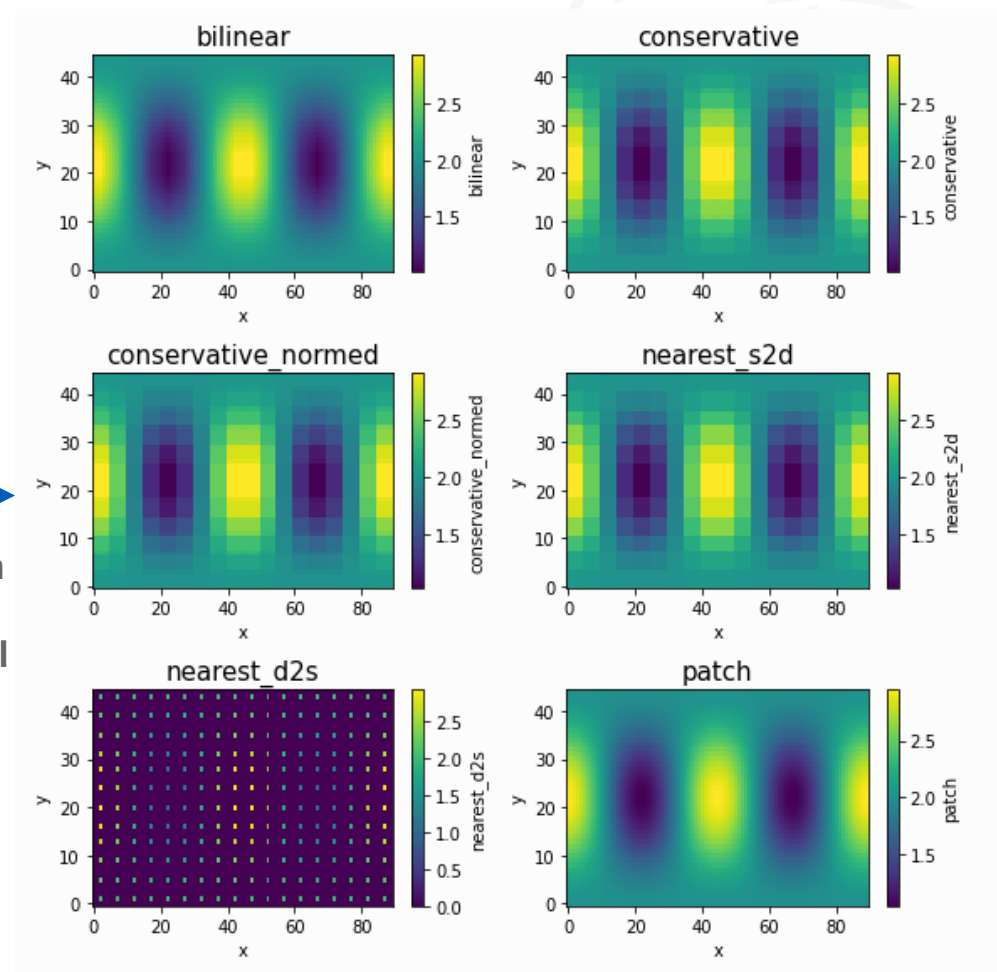
Regridding



Regrid

When the spatial resolution of meteorological forcing are different than the model domain

Six different regridding methods



Homework #7

- Visualization Battle!!

